12.3 Surface Areas of Pyramids and Cones

Name _____ Period ____ Date _____

Geometry 12.3 · Student sheet

I can…

- find the surface area of a regular pyramid.
- find the surface area of a right cone.
- find a slant height with the Pythagorean Theorem.

Vocabulary · fill in the missing word

Regular pyramid — a pyramid whose base is a regular polygon and whose lateral faces are _____ isosceles triangles

Slant height — the height of a lateral face of a regular pyramid, or the _____ from the vertex of a cone to the edge of its base

Lateral surface of a cone — the _____ surface of a cone, between the base and the vertex

Part 1 · Discovery: roll a cone

Cut a quarter out of your circle (radius 10 cm). Tape the rest into a cone. Measure the radius of the cone's base.

Base radius r	Area of the paper: ¾ · π · 10^2	π · r · 10	What is the 10 cm on the cone?

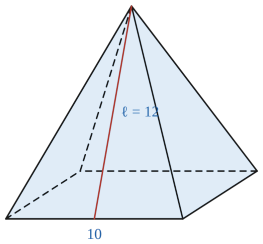
The curved surface of a cone has area π · (radius) · (_____).

Part 2 · Notes and examples (fill in each note, then work the example)

Surface area of a regular pyramid: S = B + _____, where P is the perimeter of the base and ℓ is the slant height.

Example 1

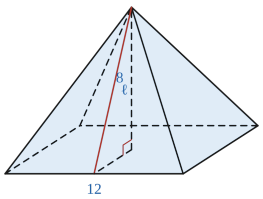
Find the surface area of the regular square pyramid.



Finding a slant height: the height, half a side of the base (or the radius), and the slant height form a _____.

Example 2

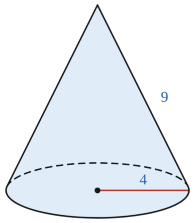
A regular square pyramid has a base edge of 12 and a height of 8. Find its slant height and its surface area.



Surface area of a right cone: S = πr² + _____

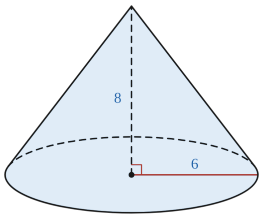
Example 3

Find the surface area of the cone. Round to the nearest hundredth.



Example 4

A cone has a radius of 6 and a height of 8. Find its surface area. Round to the nearest hundredth.



Example 5

A cone-shaped paper cup has a radius of 2 inches and a slant height of 5 inches. How much paper is in the cup (no lid)? Round to the nearest hundredth.

Part 3 · Partner practice

Partner 1

A regular square pyramid has a base edge of 8 and a slant height of 5. Find its surface area.

Partner 2

A cone has a radius of 5 and a height of 12. Find its surface area. Round to the nearest hundredth.